

AI/ML SW Integration Strategy

Presented By: Alex Seguin

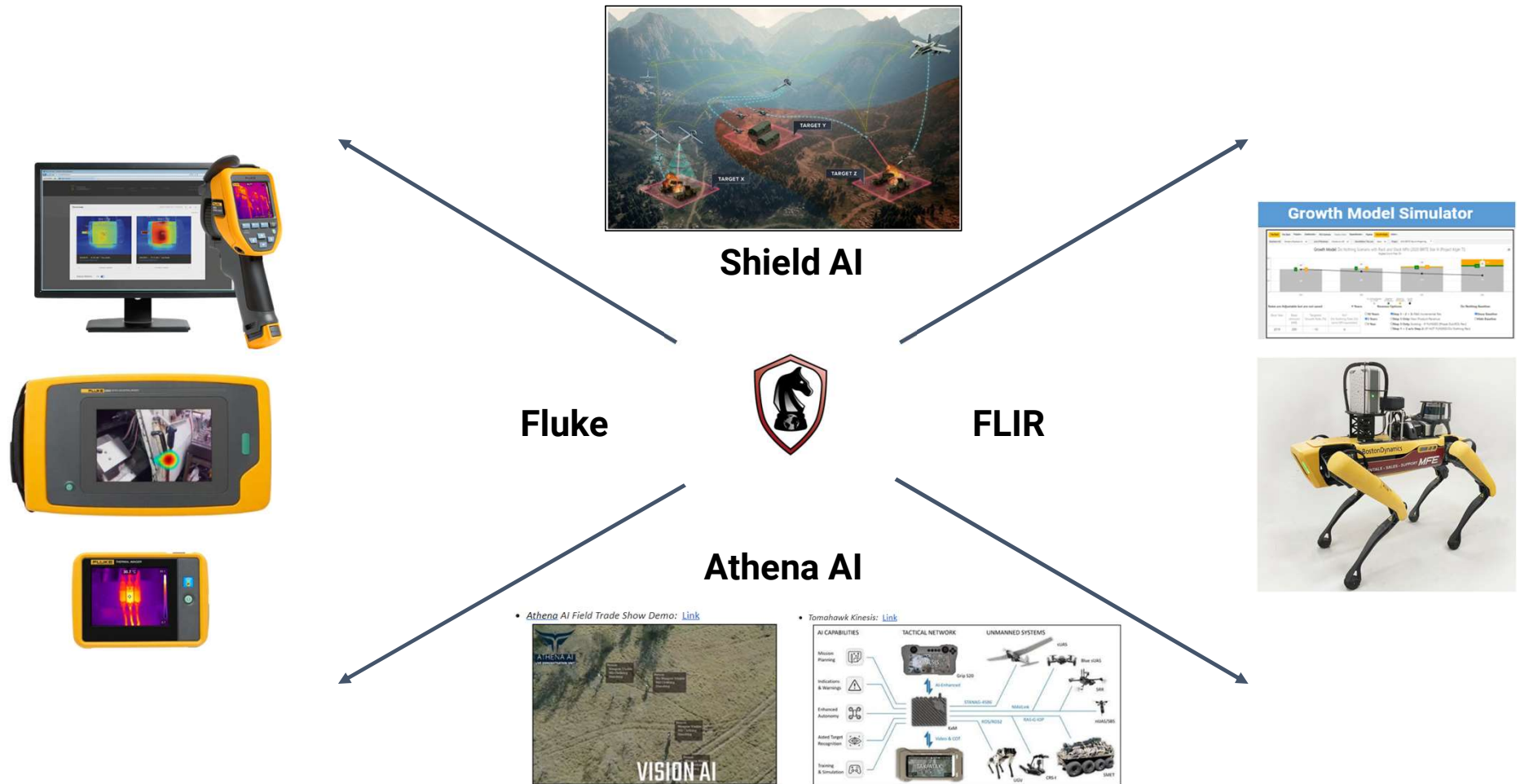
Completed: 24 JAN 2023



Experiences:

- [Strategic Marketing – Honeywell](#)
- [Industrial Imaging – Fluke](#)
- [AI EdTech – Sensor Fusion](#)
- [Portfolio Management Highlights](#)
- [Sensor Equipped Robotics](#)

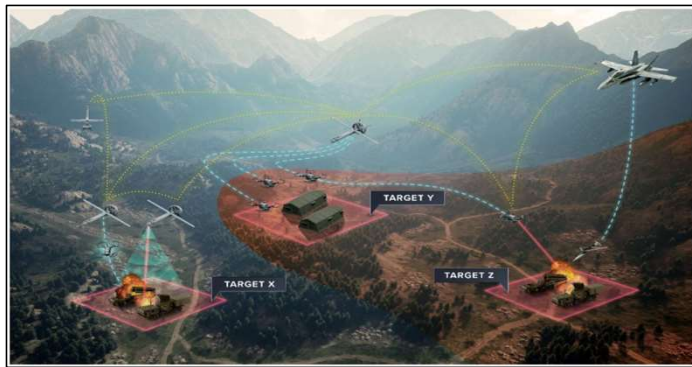
De-Risking: Growth, Strategy, and Execution



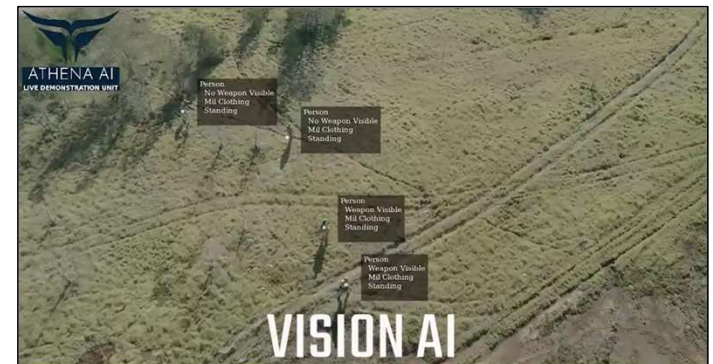
Defense Highlights: Portfolio Strategy + Business Development

[Desktop](#)

Shield AI



Athena AI



Case Study – Talking Points



DoD Needs and Benefits: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001

Think B2D, Act B2B

DoD Landscape

• **Recruiting Shortage:** Army: 15k, USAF Pilots: 1.7 k

• **Asymmetric Advantaged Desired**

- Tech + AI Mastery + Collaboration
- Autonomous UAS/UGV
- Teaming (MuM-T / HMT)
- Modular Compute Options
- Integration Ease



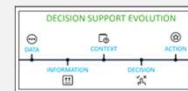
Needs & Benefits

• **Multi-Domain:** Sensor Fusion + AI



• **Joint & Allied Force Operations**

• **OODA Loop + Kill Chain Mastery**



• **Eco-System:** Targeted Innovation & Prioritization

Outcomes

• **Personas:** Field Operatives, Officers, Commanders

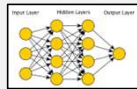
• **Metrics to Impact:**

1° Metric	Baseline	With AI	Delta
Decision Speed	120 min	45 min	75 min
Risk Score	70%	85%	15%
Effect Speed	180 min	60 min	120 min

• **Eco-System:** Interoperability + Teamwork

Execution Challenges

• **AI/ML Training**

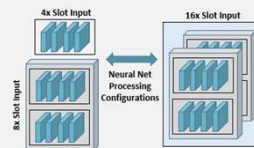


• **Simulation**

• **Modular Compute**

• **AI Agents + Auto Behaviors**

• **Eco-System:** Partnership & Reuse via Collaboration



Solution & Value Capture

• *Think B2D, Act B2B*

• **Value of Metrics:** Baseline v Implemented Outcomes

Total Impact Per Operations, M\$	\$ 2 - 5 M
Conservative Estimated Impact Per Year, M\$	\$ 125 - 150 M

* Does not include Casualties, Loss of Life, nor Loss of Military Assets

• **Eco-System:** Standard APIs + Operational Procedures

Strategy & Execution

• **Acceleration Opportunity's**

- Multi Drone + UGV + Teaming
- Enhanced Value Creation with AI
- Focus on Outcome to Understand Speed & Risk



• **Eco-System:** Multi-Party Collaborations + Legal Docs

Phased Development: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001

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Phase 1 – 2: \$6.9 M

Integration Multi-Drones with Real-Time Outputs into real-time 3D Engine

- **Ph1-TO-01:** MVP of 3x drones with existing FLIR sensors and Muve 360 MVP1.0 w/ IVAS integration
- **Ph1-TO-02:** Software MVP for UI/UX with drafted UI/UX of auto-notifications highlighting targets of interest

Multi-Domain UGV, Heavy Lift, Taser Drone with eXeL AI Edge Processing

- **Ph2-TO-03:** Integration of single GPU for AI edge compute along w/ UGV integration
- **Ph2-TO-04:** Integration of multiple AI edge compute/GPU modules + heavy lift LD UAS and custom taser drone
- **Ph2-TO-05:** IVAS integration to Tomahawk controller w/ AI edge compute, Kägwerks & In-App approvals
- **Ph2-TO-06:** Software Design with auto-notification & “In-App” real-time escalations and approvals



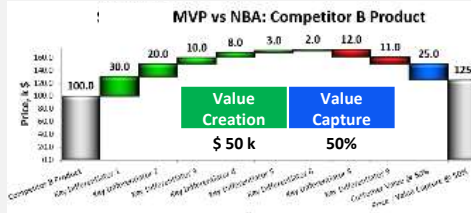
Phase 3 - 4 : \$10.6 M

Ruggedized AI Deployment of mUAS Aircrafts + Project Convergence '23-24

- **Ph3-TO-07:** Testing 01 of Alpha eXeL AI MVP1.5 with IVAS accessible approvals
- **Ph3-TO-08:** AI/ML Model Dev 01: Iterate on selected AI/ML models, AI autonomous behaviors, and use cases
- **Ph3-TO-09:** Final hardware selection and production ready enclosures and deployments
- **Ph3-TO-10:** Testing 02 of Beta eXeL AI Prototype with an MVP2.0 of edge compute and integration for JADC2

Utilizing SatCom, VBAT 228, and a Trillium AI Edge Payload for Laser Designation

- **Ph4-TO-11:** AI/ML Model Dev 02: Securely LIVE update new AI/ML models & autonomous AI behaviors
- **Ph4-TO-12:** SatCom integration of Honeywell system, and A2AD demo of MVP2.5
- **Ph4-TO-13:** Final Hardware and Software deployment with most recent Software and AI models (MVP3.0)

eXeL AI - C4				Functional Lead	Name	COGS	To-Date	2023	Life	Financials		NextGen/Refresh																										
COGS	MSRP	Mgn %		Product Mgr	F. Last	I-RAD	\$10 M	--	--	1 st Rev Date	Q# - YY	Desktop																										
\$ -- k	\$ -- k	%		Project Eng	F. Last	C-RAD	\$24 M	--	--	3-Year NPV	\$ -- M																											
				BD/Sales Lead	A. Seguin	Cap-Ex	\$ 3.4 M	--	--	IRR	%																											
Project Overview				Market Opportunity						Portfolio Positioning																												
Project Summary - eXeL AI-C4 is an AI platform ecosystem that enables commanders to <i>exploit mission-specific UxV information</i> from the field at the speed of relevance, <i>reducing risk to mission</i> and <i>improving decision making</i>. - In Q3 2024, eXeL AI-C4 seeks to enter Full Rate Production, on the STD+ MIL system variants.				Franchise Pursuits <table><thead><tr><th>Pursuit</th><th>Award</th><th>Full value</th></tr></thead><tbody><tr><td>Prog 1</td><td>2021</td><td>\$ - M</td></tr><tr><td>Prog 2</td><td>2022</td><td>\$ - M</td></tr><tr><td>Prog 3</td><td>2023</td><td>\$ - M</td></tr></tbody></table> CRM: Sales Force <table><thead><tr><th>Quarter</th><th>Award</th><th>Full value</th></tr></thead><tbody><tr><td>Q1</td><td>2023</td><td>\$ 5.0 M</td></tr><tr><td>Q2</td><td>2023</td><td>\$ 15.0 M</td></tr><tr><td>Q2</td><td>2023</td><td>\$ 2.0 M</td></tr><tr><td>Q4</td><td>2023</td><td>\$ 6.3 M</td></tr></tbody></table> CONOP - UAS01 and UGV 01 assess Site X, flagging 4 RedFor Objects of concern (A) - (D). - UAS02 is launched from 1km away at Site Y, performing risk and threat assessment at Site X and delivering taser effect to Target A with HITL approval. - UAS02 leaves to return to offsite, UAS01 assesses neutralization of Target A.						Pursuit	Award		Full value	Prog 1	2021	\$ - M	Prog 2	2022	\$ - M	Prog 3	2023	\$ - M	Quarter	Award	Full value	Q1	2023	\$ 5.0 M	Q2	2023	\$ 15.0 M	Q2	2023	\$ 2.0 M	Q4	2023	\$ 6.3 M	Price + Performance 
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Product Strategy / Value Proposition				Dependencies: Intra-BU/LOB + Customer						Roadmap + Schedule																												
Product Strategy - Autonomously execute user-defined tasks with ID, geo-location, and fusion capabilities. - Leverage AI/ML algorithms from UGVs, UASs, and aircraft video feeds. - Enable recording and playback of real-world missions with digital twinning for AI autopilot adaptive execution. Value Proposition 1. Improve Risk Assessment Quality 2. Accelerate Speed to Decision 3. Enhance Speed to Effect 4. Optimize Asset Value				Deloitte- <table><thead><tr><th>Project</th><th>Project #</th></tr></thead><tbody><tr><td>eXeL AI-UGV</td><td>.00011</td></tr><tr><td>eXeL AI-M360</td><td>.00012</td></tr><tr><td>eXeL AI-NxtGen 2</td><td>.00013</td></tr></tbody></table> DoD - Customer <table><thead><tr><th>Project</th><th>Project #</th></tr></thead><tbody><tr><td>Army IVAS</td><td>BAA.00001</td></tr><tr><td>Army Vantage</td><td>BAA.00002</td></tr><tr><td>USAF Golden Horde</td><td>BAA.00003</td></tr></tbody></table>						Project	Project #	eXeL AI-UGV	.00011	eXeL AI-M360	.00012	eXeL AI-NxtGen 2	.00013	Project	Project #	Army IVAS	BAA.00001	Army Vantage	BAA.00002	USAF Golden Horde	BAA.00003	MVP vs NBA: Competitor B Product 												
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❖ Drone ISR & System Notifications												❖ AI Behaviors to Multiple mUAS	❖ SatCom + AI/ML Model Updates	❖ Accelerate eXeL AI Eco-System																								
Q1 2023	Q2 2023	Q3 2023	Q4 2023	2024	2025	2026																																

Market Landscape + Program of Record: *Unmanned UAS / UGV*

Desktop

Business: eXeL AI Project #: 2023.P001



Application Alignment

Autonomy: Manned w/ AI, Remote control w/ AI, Unmanned - fully autonomous AI
Wingman: AI co-pilot, AI enabled training, Unmanned AI aircraft protection
Intelligence: Remote piloted aircraft with AI intelligence / ISR optimized capabilities
Defense Logistics: Manned/Unmanned AI delivery, AI inspection for maintenance
Interstellar: Space/Mars specific AI powered aerial craft. Note: Far future
Other: Air Launch Effects (ALE) + Loitering Munitions

Aircraft Integration Possibilities

Multi-sized aircraft Group 1/2/3: Quad/VTOL/Fixed, Group 4/5 integration partners, Larger Quads: Electric/Fuel/Jet, Helicopters, Ground Systems + Sea Vessels.



Programs of Record + Trade Shows

Space / Air Force

Teaming: [Golden Horde](#), [Skyborg](#)

Other: [NTS-3](#), [Rocket Cargo](#)

Army Programs

Mobility: [FTUAS](#)/[TUAS](#)/[EUAS](#), [FVL CFT](#), [NGCV](#) ([RCV](#), [OMT](#)), [CRS-H](#), [CRS-H](#), [AMPV](#), [AWS](#)

Decision: [AI2C](#), [ITAS](#), [FASIT](#), [Vantage](#), [Cyber SU](#), [SL CFT](#), [GFT](#), [ARL-M](#), [APNT](#), [N-CFT](#), [Air SS](#), [ITN](#),

Navy/Marines: MI2, [ONR](#), [AWS-UAS](#)

SOF: PEO-C4, PEO-FW, PEO SDA

DHS: [C-UAS](#)

Trade Shows

Army: AUSA

Air Force: AFA

Navy: Sea, Air Space

Marine Corps: Modern Day Marine international: DESI + Eurosatory
Non-Service Specific, Industrial OR commercial: AUVSI-Xponential, AUVSI-DPS, CES.

Publications:

Janes, Shephard, National Defense Magazine, Defense Daily, Breaking Defense



Value Creation Opportunities

Enhancements

AI-Payloads: Modular sensor add-ons, integration speed, upgradable AI for vision
Modular Compute: Ability to quickly replace & scale AI compute: Edge TPU/GPU/etc.
Propulsion Options: Gas, Heavy/Jet Fuel, Hydrogen, Batteries/Hybrid, Ion Thrusters
Custom AI Missions: Auto-data recordings, user trained AI, user optimized missions
Field AI Upgrades: Cyber-secure upgrades for new AI model testing + management
Charging Options: Low maintenance charging stations, Auto-battery replacements
Targeting Capabilities: Human + AI on moving targets w/ casualty + RISK assessment
Stealth Options: Stealth fixed/VTOL aircraft integration partners

Integrations

Teaming: Multi-UAS/UGV missions, AI enhanced insight, & Human-in-the-Loop [HITL]
AI Ranked Threats: Prioritized threats w/ Fixed, PTZ, & GRD/AIR mobile AI systems
AI Tracking / Assist: Auto-tracking – separate control of payload system
HITL Decision: AI assessed threats, actionable insight, enhanced HITL decision speed
Extra Edge Compute: Off-aircraft AI edge compute or mobile Edge drone
Remote Console: Command/control center, cyber-secure platform APIs, duel-view
AR/VR: Display integrations, with optimized AI/ML with controls via voice/touch/etc.

Deployment:

Ground: Human: Manual/Semi-auto deploy, Auto-deploy: stationary + vehicles + box
Aerial: Aerial deployment, Booster/rocket aerial assist
Water: Ocean/Lake vessel, Sub-sea to aerial

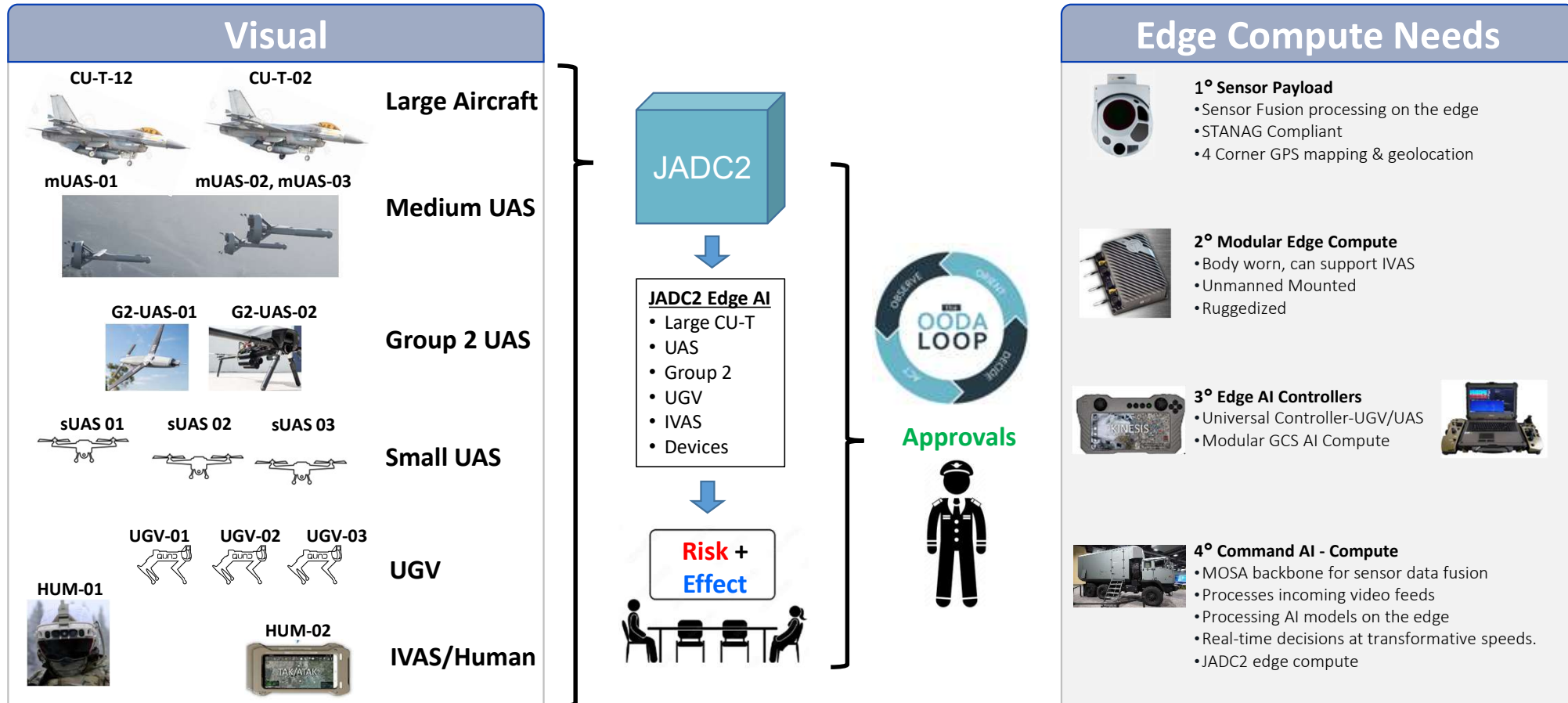
Recovery

Ground: Human, Ground mobility system: Remote + Autonomous AI recovery
Aerial: Pickup Day Drone, Pickup Night Drone
Damaged: Field destroyed field unit: SW/AI memory wipe + deploy HW interlocks.

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Multi-Domain + Edge Compute: *Landscape*

Business: eXeL AI Project #: 2023.P001



DoD Needs and Benefits: *eXeL AI - C4*

Business: -eXeL AI Project #: 2023.P001

Think B2D, Act B2B

Benefits

- **Reduced Risk:** Reduce Risk to Force and Mission with AI add-ons
- **Decision Support:** Enhance existing assets capabilities with AI Object Detection
- **Adaptability:** ATAK integration with auto-notifications to Battle Field Commanders
- **Multi Domain:** Interoperability to switch easily between UGV robotics and UAS drones
- **Outcome Focused:** Improved speed to decision & speed to effect

Primary Metrics

- **Time to Decision**
- **Risk Assessment**
- **Time to Effect**

1 st Metric	Baseline	With AI	Delta	Value Est* Per Op
Decision Speed	120 min	45 min	75 min	\$ 2.0 - 5.0 M
Risk Score	70%	85%	15%	
Effect Speed	180 min	60 min	120 min	

Additional Metrics

Secondary Metrics: Resourcing

- Agents + BlueFor Tracking
- Risk Analysis of Threats
- Reviewers + Support for Decisions
- Approvals + Commanders

2 nd Metric	Baseline	With AI	Delta	Value Est* Per Op
# Agents	5	4	4	\$ 0.1 M
# Analysts	10%	6%	6%	
# Reviewers	3	2	2	
# Approvers	2	1	1	

Tertiary Metrics: Optimization

- Value of Physical Assets
- # Data Handoffs
- Bandwidth Data Limitations
- Total Data Transferred

3 rd Metric	Baseline	With AI	Delta	Value Est* Per Op
Assets Deployed, M\$	\$ 150 M	\$ 110 M	\$ 110 M	\$ 0.150 M
# Data Handoffs	100	50	50	
# Data Transferred	500 GB	100	100	
Max Data Bandwidth	10 GB / sec	10 GB / sec	10 GB / sec	

Valuation Per Operation

1 st Metric	Delta	Value Est* Per Op
Decision Speed	75 min	\$ 5.0 M
Risk Score	15%	
Effect Speed	120 min	

2 nd Metric	Delta	Value Est* Per Op
# Agents	4	\$ 0.100 M
# Analysts	6%	
# Reviewers	2	
# Approvers	1	

3 rd Metric	Delta	Value Est* Per Op
Assets Deployed, M\$	\$ 110 M	\$ 0.150 M
# Data Handoffs	50	
# Data Transferred	100	
Max Data Bandwidth	10 GB / sec	

Total Impact Per Operations, M\$	\$ 5.2 M
Estimated Impact Per Year, M\$	\$ 125 - 150 M

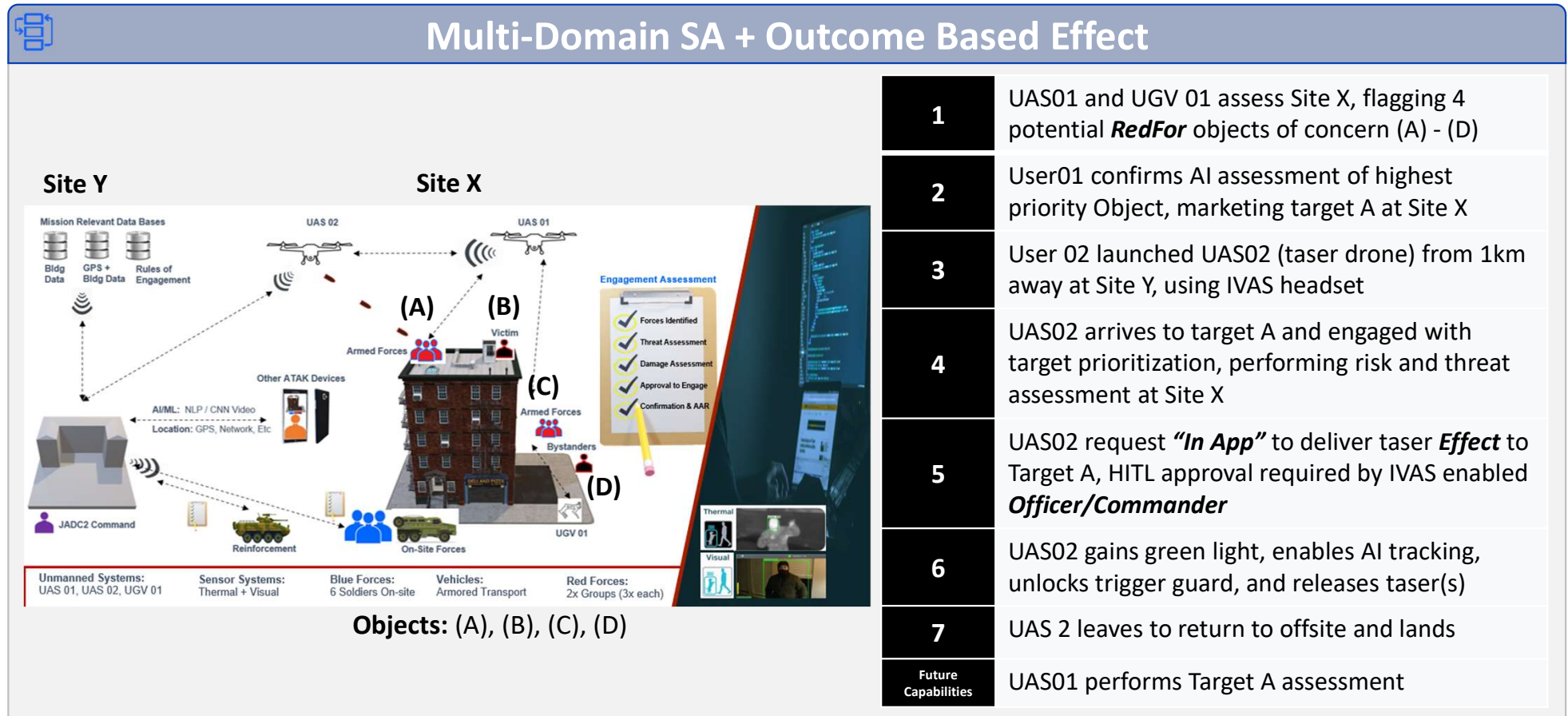
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DRAFT - CONOPS: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001



Customer Problem Statement: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001

1	Problem Statement	Public Safety and Defense Officers involved in a dynamic time-sensitive targeting situation need faster intelligence, accurate target acquisition, automation of manual work to enable quicker and safer decision making
2	Customer Outcome	- Accuracy in locking onto target (active shooter - armed) - Faster and accurate risk calculations - Better visual situational awareness (SA) of threats and risks - Execution of Taser Drone to neutralize target with non-lethal force + “In App” confirmation
3	Customer Why's	- No CIVCAS - eliminate risk to SA & collateral damage - Automation of multiple sources of data to reduce human error - Need higher confidence to decide prior to delivery of effect
4	Problems with the Status Quo	- Potential for human error is high (manual calculations, maintaining lock on moving target) - Current software can be cumbersome and lagging, not built to automate - Not enough people resources to support multiple ops happening simultaneously

Execution Challenges: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001

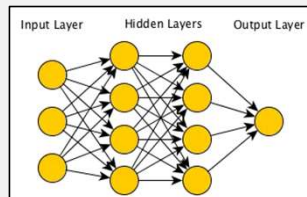
AI/ML Training & Scalability

Knowhow and Data Security

- Sensor Expertise
- Annotation
- Augmentation
- Auto Labeling

AI/ML Expertise and Optimization

- ML Researchers
- ML Dev / Ops
- ML Consultants



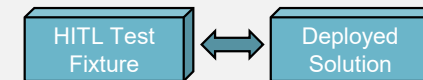
Simulation Needs

Acceleration methods for AI/ML Hardware



Accelerate DEVELOPMENT SPEED → LIVE DEMO

- Edge Intelligence
- Human + Unmanned



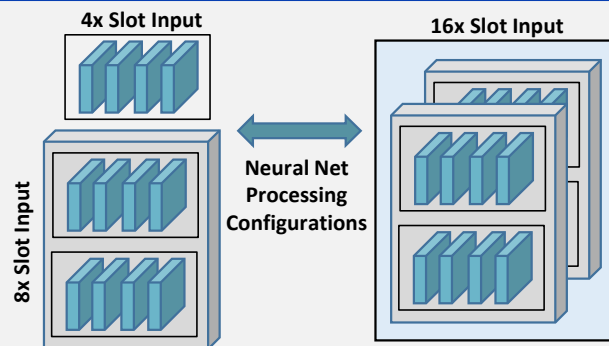
Modular Edge Compute Needs

Knowhow and Data Security

- Sensor Expertise
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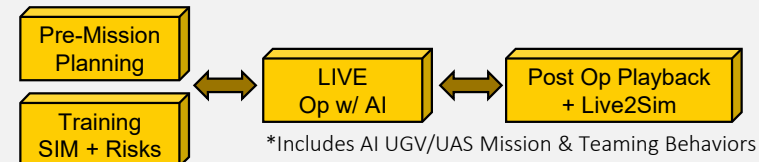


AI Agents and Behaviors

Today: Manual & Disconnected Processes, with Limited IT Connectivity to AI Enabled Systems

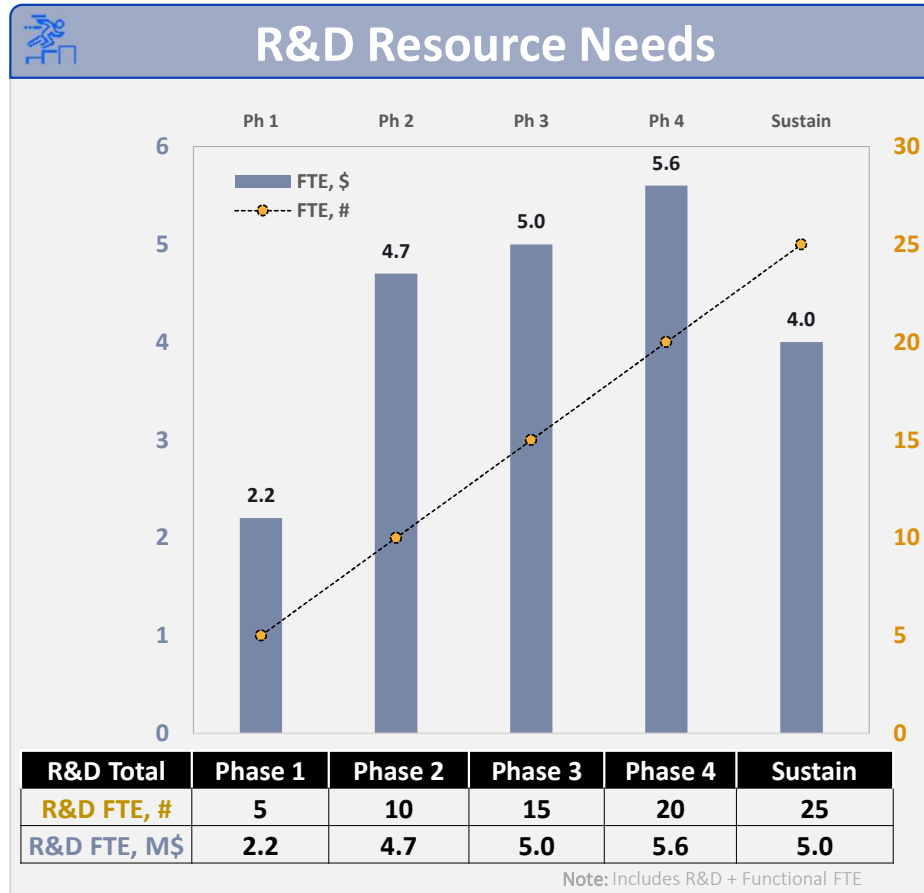


Future: Automated & Connected Processes, with Abundant Connected AI Enabled Systems



Functional Resource Alignment: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001



Functional Alignment By Quarter

Total, FTE #	Q1-23	Q2-23	Q3-23	Q4-23	Q1-24	Q2-24	Q3-24
R&D - Engineering	# 3.0	# 5.0	# 7.0	# 11.0	# 14.0	# 14.0	# 12.0
R&D - UX / ID	# 0.5	# 1.0	# 2.0	# 2.0	# 3.0	# 3.0	# 2.0
Mkt / Sales / BD	# 0.5	# 2.0	# 3.0	# 3.0	# 4.0	# 5.0	# 4.0
Dev Ops / Supply	# 1.0	# 1.0	# 2.0	# 2.5	# 3.0	# 3.5	# 3.5
CS /Support / Service	# ---	# 1.0	# 1.0	# 1.5	# 2.5	# 3.0	# 3.5

- **Technical + Manufacturing Resources**
 - Largely software and AI developers, with expert UI/US for mission approval process
 - Hardware + Firm-ware expertise, largely to work with robotics system integrations
- **Commercial Resources**
 - Minimal marketing additions for lead generation, and speaking engagements
 - Business development and sales, to follow-up opportunities from leadership teams
- **Supporting Infrastructure**
 - Dev Ops and ML Ops to continue internal process alignment to release additional capabilities
 - Support members to trouble shoot and follow-up on customer issues after deployment

FTE k\$ Estimate	Q1-23	Q2-23	Q3-23	Q4-23	Q1-24	Q2-24	Q3-24
Marketing / Sales	75	300	450	450	600	750	450
Dev Ops / Supply	125	125	250	313	375	438	438
Support / Service	---	100	100	150	250	300	350

R&D Assets Only: *eXeL AI - C4*

Business: eXeL AI Project #: 2023.P001



Phase 1 – 2: \$1.6 M

•Integration Multi-Drones with Real-Time Outputs into real-time 3D Engine

Phase	Hardware	Qty	Budget
1	Teal 2	2	35
	Hadron 640R	6	25
	FLIR Drone	1	25
	Tomahawk Kinesis	2	30
	Unreal Model	1	5
	Unreal Assets	10	10
	Contingency - Addn'l Needs	1	70
			200

•Multi-Domain UGV, Heavy Lift, Taser Drone with eXeL AI Edge Processing

Phase	Hardware	Qty	Budget
2	Ghost Robotics Vision 60	1	160
	Axon Taser	1	2
	Custom Drone + Mount	1	20
	Heavy Lift	1	80
	Laser Designation Payload	1	200
	GPU - Non-Rugged	6	30
	Contingency - Addn'l Needs	1	208
			700



Phase 3 - 4 : \$3.1 M

•Ruggedized AI Deployment of mUAS Aircrafts + Project Convergence '23-24

Phase	Hardware	Qty	Budget
3	Ruggedized GPU Compute	1	120
	mUas Quantum Systems	1	25
	Payload - Trillium HD95-LD	1	500
	mUAS AeroVironment	1	800
	Contingency - Addn'l Needs	1	55
			1,500

•Utilizing SatCom, VBAT 228, and a Trillium AI Edge Payload for Laser Designation

Phase	Hardware	Qty	Budget
4	mUAS - VBAT 228	1	1,200
	Honeywell - SatCom	1	200
	Contingency - Addn'l Needs	1	200
			1,600

Strategy Development & Execution: **eXeL AI - C4**

Business: eXeL AI Project #: 2023.P001

3 Year Objectives	2023 Objectives	Targets to Improve
- Grow from a 2022 baseline global contribution margins to 20x margins by 2026 - Increase from a 2022 baseline of X to 10X the number of Collaborative Partnerships to accelerate development by 2026 - Grow from 2022 baseline of Y to 10Y minority investment positions & inorganic M&A screening opportunities by 2026	- Grow 2022 baseline USA contribution margins to 10x 2022 margins by 2023 - Grow 2022 baseline APAC + EMEA contribution margins to 5x 2022 margins by 2023 - Accelerate the Collaborative Partnerships development by increasing 2022 funnel entry of X to 5X for 2023 - Increase 2022 baseline of # of minority investment positions from Y to 3Y by 2023 - Increase 2022 # of inorganic M&A screening from Y to 3Y and # of M&A acquisitions from Y to 2Y by 2023	- TTI #1 Increase # of USA based engagements to 15X of 2022 revenues and 10x of 2022 margins by December 15, 2023 - TTI #2 Increase # of APAC and EMEA based engagements to 15X of 2022 revenues and 10x of 2022 margins by December 15, 2023 - TTI #3 Increase the Win/Loss rate by 15% for global customer engagements by December 1, 2023. - TTI #5 Accelerate the lead generation of Collaborative Partnerships by increasing 2022 funnel entry from X to 10X by December 15, 2023 - TTI #6 Reduce the initial project execution period of Collaborative Partnerships by 50% of 2022 timelines by December 15, 2023 - TTI #7 Increase the 2022 # of minority investment positions from to 2Y and 3Z dollars invested by November 3, 2023. - TTI #8 Increase the 2022 # of inorganic M&A screening reports to 3Y by November 10, 2023. - TTI #9 Increase the 2022 # of M&A acquisitions to 2Y by December 31, 2023.

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